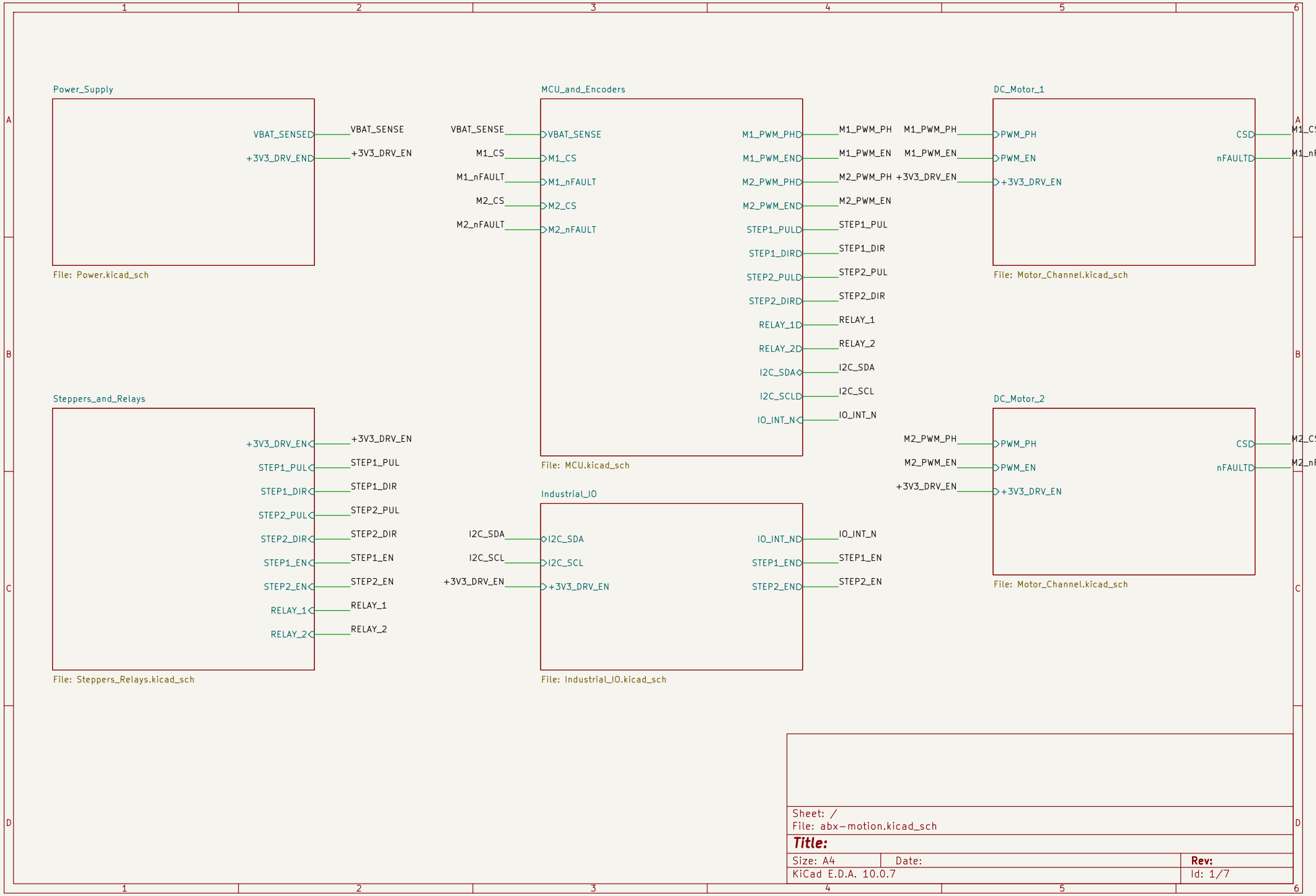
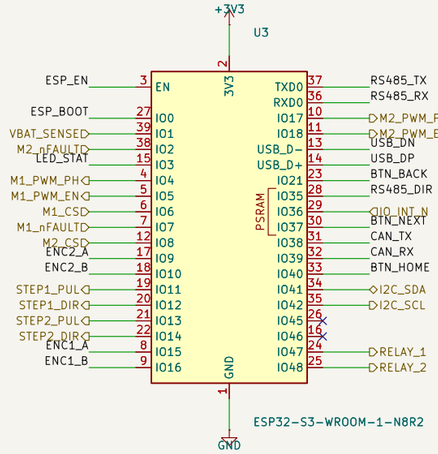

Power_Supply
VBAT_SENSE
+3V3_DRV_EN
VBAT_SENSE
+3V3_DRV_EN
File: Power.kicad_sch
MCU_and_Encoders
VBAT_SENSE
M1_CS
M1_nFAULT
M2_CS
M2_nFAULT
M1_PWM_PH
M1_PWM_EN
M2_PWM_PH
M2_PWM_EN
STEP1_PUL
STEP1_DIR
STEP2_PUL
STEP2_DIR
RELAY_1
RELAY_2
I2C_SDA
I2C_SCL
IO_INT_N
File: MCU.kicad_sch
DC_Motor_1
PWM_PH
PWM_EN
+3V3_DRV_EN
CSD
nFAULTD
M1_CS
M1_nFAULT
File: Motor_Channel.kicad_sch
Steppers_and_Relays
+3V3_DRV_EN
STEP1_PUL
STEP1_DIR
STEP2_PUL
STEP2_DIR
STEP1_EN
STEP2_EN
RELAY_1
RELAY_2
File: Steppers_Relays.kicad_sch
Industrial_IO
I2C_SDA
I2C_SCL
+3V3_DRV_EN
IO_INT_ND
STEP1_END
STEP2_END
File: Industrial_IO.kicad_sch
DC_Motor_2
M2_CS
M2_nFAULT
File: Motor_Channel.kicad_sch
Sheet: /
File: abx-motion.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 1/7

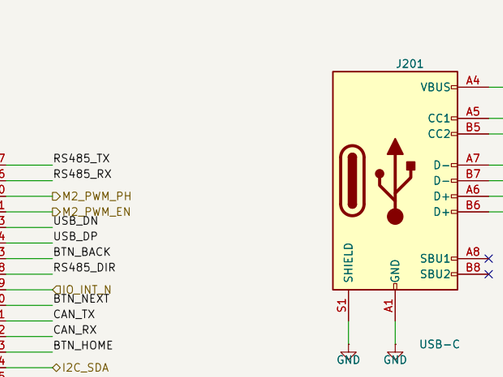
SHEET 1: BATTERY INPUT, REVERSE POLARITY PROTECTION, 5V BUCK & 3.3V LDO

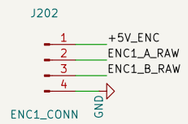
+BATT
PWR_FLAG
+5V
PWR_FLAG
PWR_FLAG
GND
J1
XT60_BAT
V_RAW_IN
GND
F1
25A
V_FUSED
Q1_GATE
Q1
SQJ459EP-T1_GE3
V_FUSED
+BATT
D2
TVS3300DRVR
+BATT
GND
GND
R2
150k
VBAT_SENSE
R3
10k
GND
VBAT_SENSE
C1
100nF
GND
U1
LMR36520FADDA
+BATT
U1_VCC
U1_BOOT
U1_SW
U1_FB
GND
U1_SW
L1
15uH
+5V
D2: 33V standoff; 40V max at specified 8/20us pulse. No continuous braking—energy rating.
+BATT
BZT52C12S01
Q1_GATE
R1
47k
GND
+BATT
C2
4.7uF
GND
+BATT
C3
4.7uF
GND
+BATT
C4
100nF
GND
U1_VCC
C5
1uF
GND
U1_BOOT
C6
100nF
U1_SW
+5V
R4
100k
U1_FB
R5
24.9k
GND
+5V
C7
22uF
GND
+5V
C8
22uF
GND
U2
AP2112K-3.3
+5V
VIN
EN
VOUT
GND
+3V3
GND
+5V
C9
1uF
GND
+3V3
C10
10uF
GND
+BATT
R6
22k
BATT_LED_A
LED_GRN_BATT1
GND
+5V
R7
1k
5V_LED_A
LED_GRN_5V1
GND
+3V3
R8
1k
3V3_LED_A
LED_GRN_3V3_1
GND
J102
+3V3
+3V3_DRV_EN
E_STOP
+3V3_DRV_EN
R9
10k
GND
TP_GND1
GND
TP_GND1
TP_GND2
GND
TP_GND2
TP_BATT1
+BATT
TP_BATT1
TP_5V1
+5V
TP_5V1
TP_3V3_1
+3V3
TP_3V3_1
LOGO1
ABX_Logo_Gears_Gold
H1
GND
M3_Mount
H2
GND
M3_Mount
H3
GND
M3_Mount
H4
GND
M3_Mount
LOGO2
ABX_Text_Gold

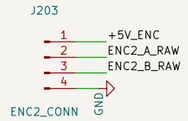
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File: Power.kicad_sch
Title:
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 2/7

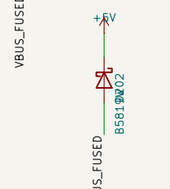
SHEET 2: ESP32-S3-WROOM-1, USB-C, 74LVC14A ENCODERS, CAN, RS-485, OLED & UI BUTTONS

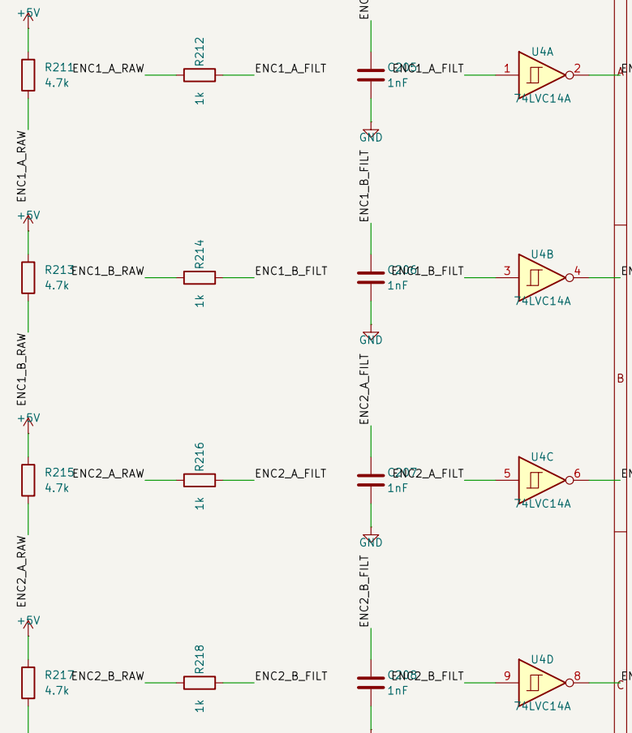


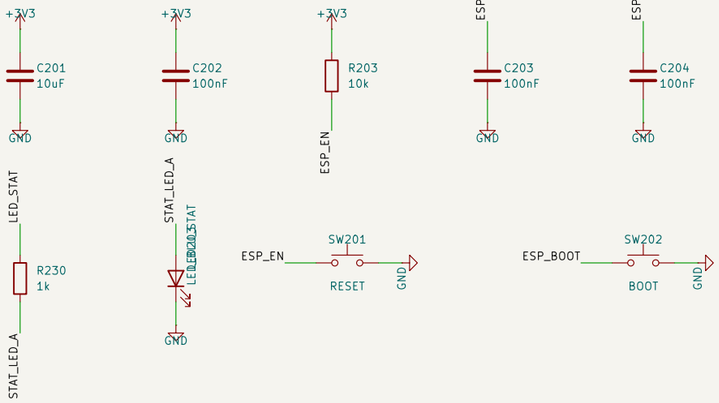


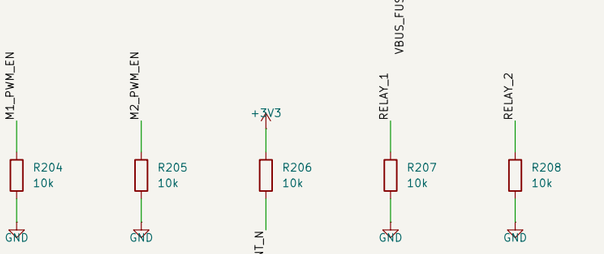


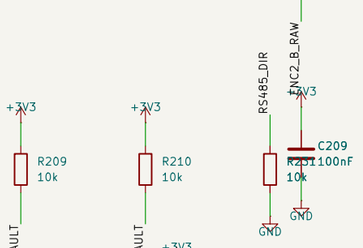


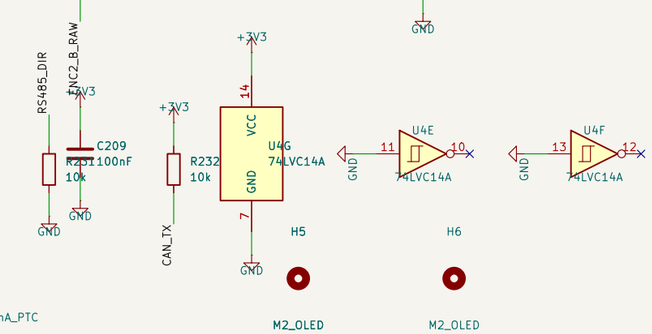


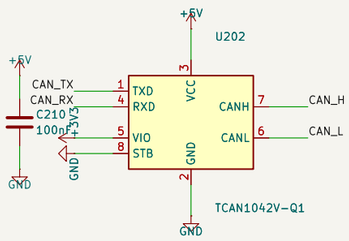


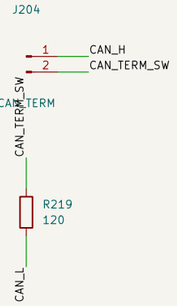


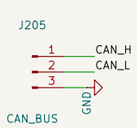




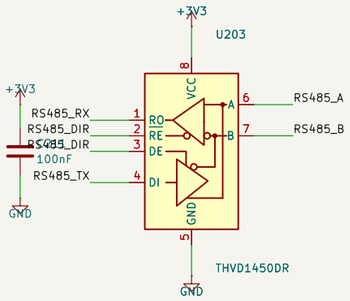


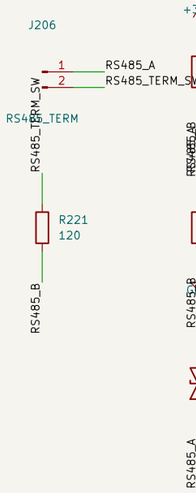


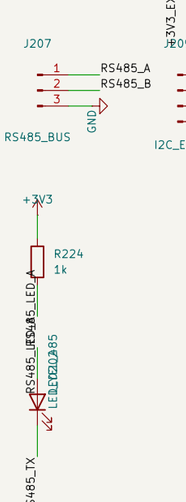


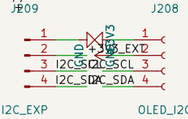


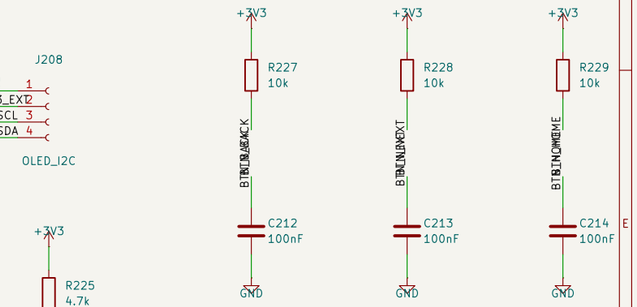


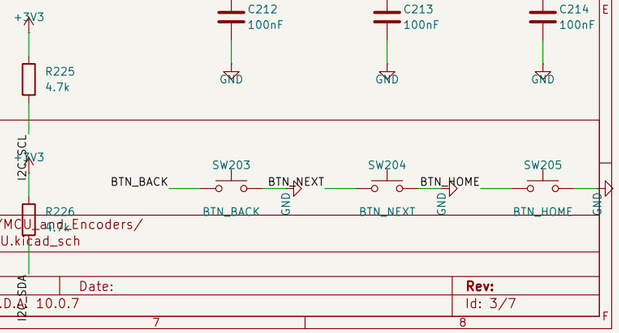










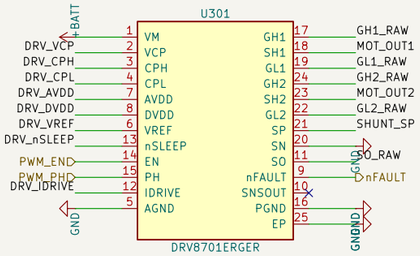


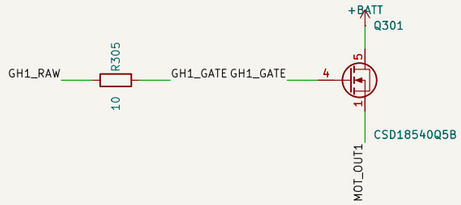
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File: MCU.kicad_sch

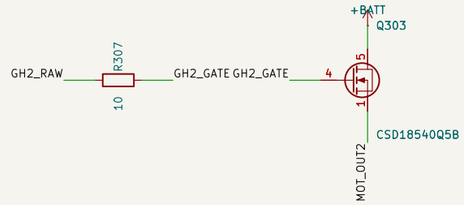
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/7

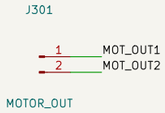
SHEET 3: 100W BRUSHED DC SERVO CHANNEL (DRV8701ERGER + 4x CSD18540Q5B 60V H-BRIDGE)











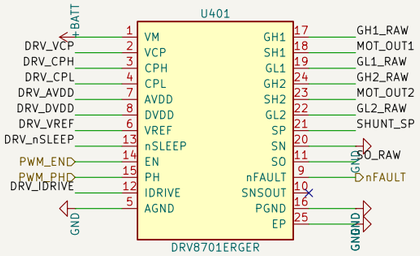


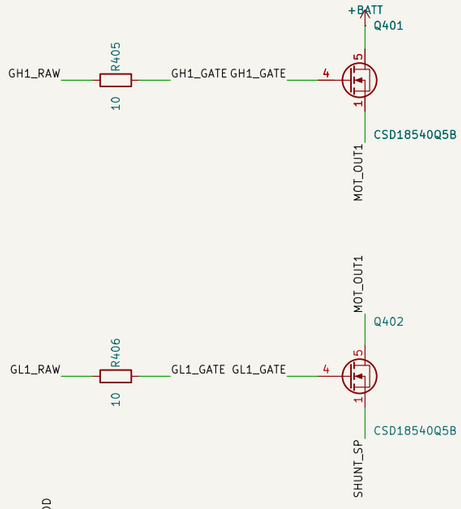


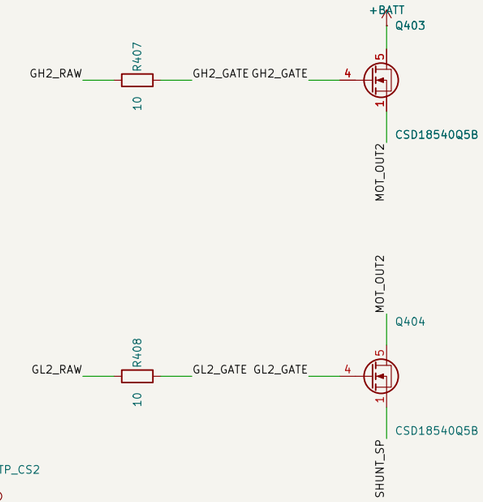
ADC protection: R304/R404 + R313/R413 divide SO by 2; rescale firmware.

Sheet: /DC_Motor_1/		
File: Motor_Channel.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 4/7

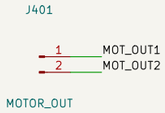
SHEET 3: 100W BRUSHED DC SERVO CHANNEL (DRV8701ERGER + 4x CSD18540Q5B 60V H-BRIDGE)





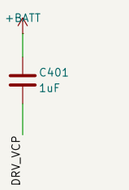




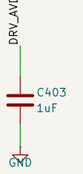








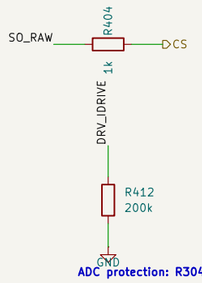




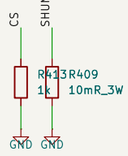
















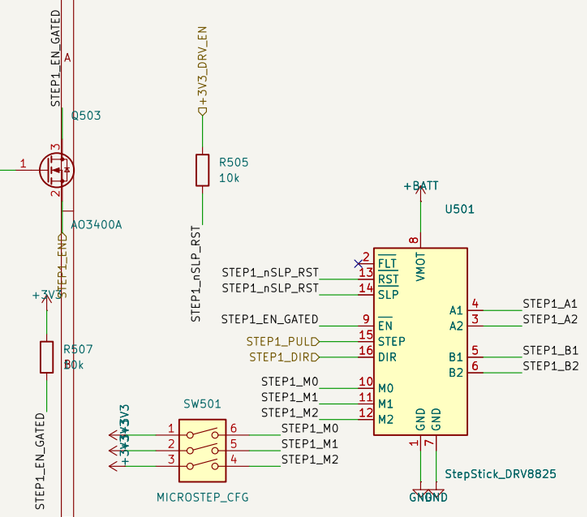


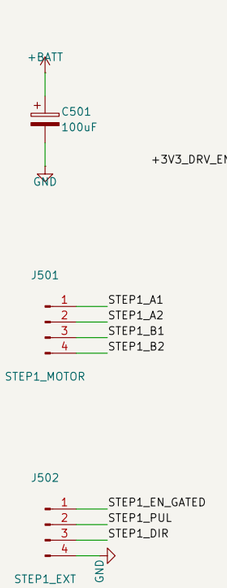


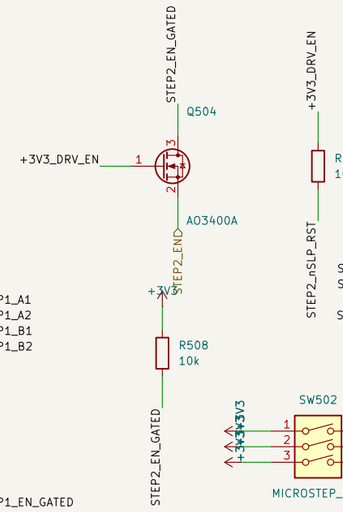
ADC protection: R304/R404 + R313/R413 divide SO by 2; rescale firmware.

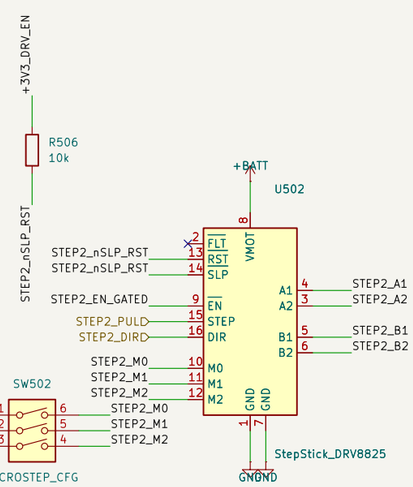
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File: Motor_Channel.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 5/7

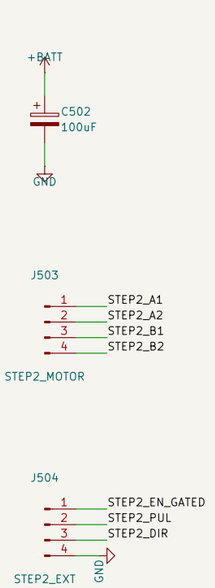
SHEET 4: DUAL STEPSTICK STEPPER CHANNELS & DUAL 5V OMRON G5LE RELAY OUTPUTS

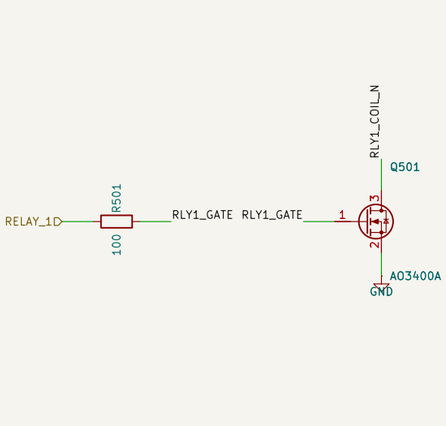




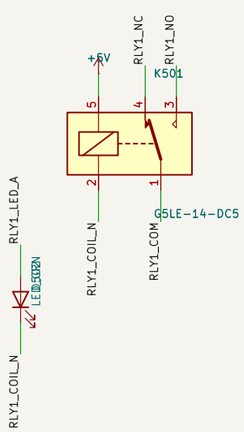


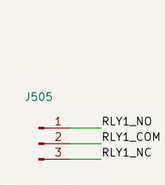


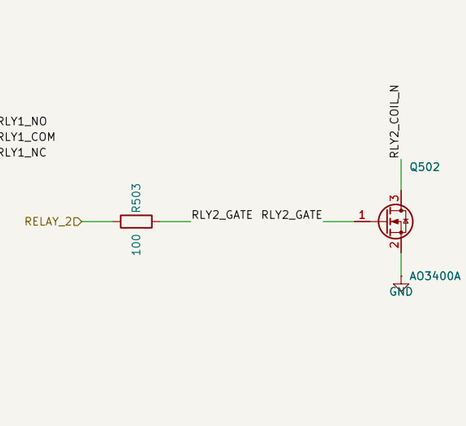




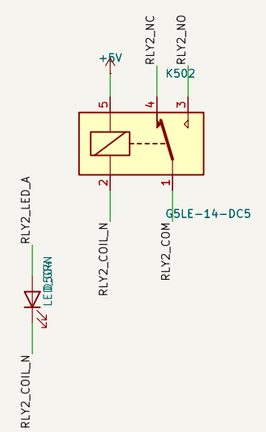


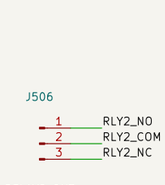












Sheet: /Steppers_and_Relays/
File: Steppers_Relays.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 6/7

SHEET 5: 6x 3-PIN JST-PH INDUSTRIAL I/O (8x LIMIT/GPIO + 2x 5V/500mA DC OUTPUTS) & PCA9555PW EXPANDER

J601
M1_MIN_RAW
M1_MAX_RAW
GND
B3B-PH-K-S

J602
M2_MIN_RAW
M2_MAX_RAW
GND
B3B-PH-K-S

J603
S1_MIN_RAW
S1_MAX_RAW
GND
B3B-PH-K-S

J604
S2_MIN_RAW
S2_MAX_RAW
GND
B3B-PH-K-S

+3V3
R601 4.7k
R602 4.7k
R603 4.7k
R604 4.7k
R605 4.7k
R606 4.7k
R607 4.7k
R608 4.7k
M1_MIN_RAW
M1_MAX_RAW
M2_MIN_RAW
M2_MAX_RAW
S1_MIN_RAW
S1_MAX_RAW
S2_MIN_RAW
S2_MAX_RAW

R609 1k
R610 1k
R611 1k
R612 1k
R613 1k
R614 1k
R615 1k
R616 1k
M1_MIN_RAW → M1_MIN_IN
M1_MAX_RAW → M1_MAX_IN
M2_MIN_RAW → M2_MIN_IN
M2_MAX_RAW → M2_MAX_IN
S1_MIN_RAW → S1_MIN_IN
S1_MAX_RAW → S1_MAX_IN
S2_MIN_RAW → S2_MIN_IN
S2_MAX_RAW → S2_MAX_IN

C602 10nF
C603 10nF
C604 10nF
C605 10nF
C606 10nF
C607 10nF
C608 10nF
C609 10nF
M1_MIN_IN
M1_MAX_IN
M2_MIN_IN
M2_MAX_IN
S1_MIN_IN
S1_MAX_IN
S2_MIN_IN
S2_MAX_IN
GND

U601
+3V3
C601 100nF
I2C_SCLD
I2C_SDA
IO_INT_N
SCL
SDA
INT
VDD
A0
A1
A2
VSS
IO0_0 M1_MIN_IN
IO0_1 M1_MAX_IN
IO0_2 M2_MIN_IN
IO0_3 M2_MAX_IN
IO0_4 S1_MIN_IN
IO0_5 S1_MAX_IN
IO0_6 S2_MIN_IN
IO0_7 S2_MAX_IN
IO1_0 OUT1_CMD
IO1_1 OUT2_CMD
IO1_2 STEP1_EN
IO1_3 STEP2_EN
IO1_4 +3V3_DRV_EN
IO1_5
IO1_6
IO1_7
PCA9555PW,118

+5V
1206L050/63YR
+5V_OUT

OUT1_CMD
R617 1k
R619 100
OUT1_GATE
Q601
OUT1_SW
AO3400A
B5819W
+5V_OUT
OUT1_SW
J605
+5V_OUT
OUT1_SW
B3B-PH-K-S

OUT2_CMD
R618 1k
R620 100
OUT2_GATE
Q602
OUT2_SW
AO3400A
B5819W
+5V_OUT
OUT2_SW
J606
+5V_OUT
OUT2_SW
B3B-PH-K-S

R621 10k
R622 10k
STEP1_EN
STEP2_EN

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